

SULIT



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

SCHOOL OF COMPUTING
Faculty of Engineering

UNIVERSITI TEKNOLOGI MALAYSIA
FINAL EXAMINATION SEMESTER II 2021/2022

SUBJECT CODE : SECR2043
SUBJECT NAME : OPERATING SYSTEM
SECTION : SECR/J/B/V/P/H / SCSR/J/B/V/P/H
TIME : PAPER 1 – 90 MINUTES
DATE/DAY : 18 JULY 2022 (MONDAY)
VENUES : Online

INSTRUCTIONS:

Answer all questions from Paper 1 and Paper 2. Show all your works.

This test will contribute 35% towards the total marks of 100%.

(Please Write Your Lecturer Name and Section in Your Answer Booklet)

Name	
I/C No.	
Year / Course	
Section (Circle)	01 / 02 / 03 / 04 / 05 / 06 / 07 / 08 / 09 / 15
Lecturer Name (Circle)	Dr Zuriahati / Mrs Rashidah Dr Nur Haliza / Dr Nor Shahida

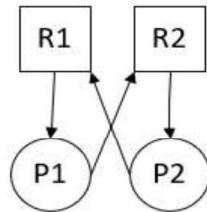
This question paper consists of 4 printed pages excluding this page.

QUESTION 1 [25 MARKS]

- a) Refer to the following set of edges that consists of request edges (R) and assignment edges (P). Answer the following questions.

$$E = \{P_1 \rightarrow R_1, R_1 \rightarrow P_2, R_1 \rightarrow P_5, P_5 \rightarrow R_2, R_2 \rightarrow P_3, P_3 \rightarrow R_3, R_3 \rightarrow P_2, R_3 \rightarrow P_1, R_3 \rightarrow P_4\}$$

- By assuming R_1, R_2 , and R_3 has 3, 3 and 1 instance/s respectively. Draw a complete Resource-Allocation Graph (RAG). Label all components clearly including the symbol '•' to represent an instance/s. [4]
 - Based on the RAG from Question 1(a), state whether the system is in a deadlock state or not. Justify your answer and circle (if any) on your RAG from Question 1(a) to denote the deadlock occurrence. [2]
 - If P_4 , has released the R_3 , what happened to the deadlock state of the system? Justify your answer your answer. [2]
- b) Refer to the following RAG, construct its corresponding **wait-for** graph to detect whether the system is in a deadlock state or not. State your conclusion? [3]



- c) Refer to Table 1 and answer the following questions. Consider the Available = (3, 3, 2, 1).

Table 1

	Allocation				Maximum			
	<i>A</i>	<i>B</i>	<i>C</i>	<i>D</i>	<i>A</i>	<i>B</i>	<i>C</i>	<i>D</i>
<i>P₀</i>	2	0	1	0	5	0	5	2
<i>P₁</i>	2	1	0	1	2	3	1	5
<i>P₂</i>	3	2	1	2	5	3	2	3
<i>P₃</i>	0	3	2	1	2	5	5	4

- By employing Resource Request Algorithm, identify whether a request from P_3 that arrives for (1, 2, 0, 0) can be granted immediately or not? If yes, write the safe sequence. Show all your work. [12]
- How many totals of instances for each of the resource A, B, C and D? [2]

QUESTION 2 [25 MARKS]

- a) A paging system with 512 pages of logical address space, a page size of 2^8 and number of frames is 1024. Answer the following questions (in bytes or in form 2^n) and show all your work.

Page number	Frame number
0	100
1	50
2	528
3	10
4	79
5	644
6	21
7	301

- What is the total size of the logical address space? [2]
- What is the total size of the physical address space? [2]
- What is the frame size of the physical memory? [1]
- How many frames of the physical memory in a number of the form 2^n ? [1]
- How many number of entry in the page table? [1]
- What is the number of bit for the page number in logical memory address and frame number in physical memory address? [2]
- What is the logical address (in decimal) given a physical address of 100001000010_2 . [2]

- b) There are 4 jobs (in order) of *A* (100 KB), *B* (350 KB), *C* (450 KB) and *D* (220 KB) need to be placed in the memory. Assume a fixed partition is used with six memory partitions of 200 KB, 300 KB, 125 KB, 600 KB, 450 KB and 250 KB (in order).
- Use the Best-fit algorithm to indicate which memory blocks are allocated to each of the four arriving jobs. Draw the memory stack and label the partition size clearly (shaded for unused). [4]
 - Does internal fragmentation occur? If yes, calculate the size of internal fragmentation before merging? [2]
- c) Assume a dynamic or variable partition is used. The following events happened:
- t_1 : process A and D left.
- t_2 : process F (550 KB) enters.

Based on **question 2b(i) answer**, draw and update the event of the memory stack at each t_1 and t_2 , and label the sizes clearly for all jobs. Use the **Best-fit algorithm and relocatable dynamic partitions** of merging the holes to place the process in the memory (shaded for unused). [8]

- t_0 : _____ and t_1 : _____
- t_1 : _____
- t_1 : _____
- t_2 : _____